# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-ELLIPSOID-EN · v1.4 · 2026-10-20 |
| Package | star-ellipsoid |
| Version | 0.1.0 |
| Capability | star.geodesy.ellipsoid |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_ellipsoid |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_ellipsoid-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_ellipsoid-0.1.0-py3-none-any.whl | 9e604b0d5adc350de0215e021f902c13598554cfdb22901e8b8d24431f42cfa0 |
| star_ellipsoid.py | 83607f7ebe9c08fb230906abcd647228f41034490cd9bb0789bb60120c637ba5 |
| star_ellipsoid-0.1.0-source.zip | 23d5309878ebca75ca5ae1c0c48c76c6b99e2906cb65fbb45433aa4de707210a |

## 5. What it does, and its limits

Radii of curvature, geocentric radius, geocentric / parametric / rectifying latitude and meridian arc on WGS-84 at a geodetic latitude; agrees with pymap3d and PyGeodesy to rounding (arc within 4e-9 m) and refuses latitudes outside [-90, 90].

The ellipsoid, not the Earth: no geoid, no terrain height. WGS-84 only (GRS80 differs in the tenth digit of f).

- `meridian_radius`, `prime_vertical_radius`, `gaussian_radius`, `geocentric_radius` (metres) at a geodetic latitude in degrees; WGS-84 constants as published.
- `geocentric_latitude`, `parametric_latitude`, `rectifying_latitude` (degrees); `meridian_arc` (metres from the equator, negative in the south; Helmert's series to n^5) and `MERIDIAN_QUADRANT` = 10 001 965.729 m.
- Measured on 505 latitudes against pymap3d and PyGeodesy: radii within 5e-16 relative, auxiliary latitudes within 5e-14 deg, meridian arc within 4e-9 m.
- Every function returns a finite float or raises `ValueError`: non-numeric, boolean, NaN or infinite input, and any latitude outside [-90, 90] (not wrapped).

## 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

# 7. Citation and bill of materials

```
@software{cavazzini_star_ellipsoid_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-ellipsoid}: The WGS-84 ellipsoid as a surface},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_ellipsoid/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts